

InSAR Time Series Subsidence Analysis

A10 Highway, Netherlands

Sloter Dijk → Sloter Vaart · January 1 – May 24, 2024
Technical Workflow Documentation and Interpretation Report

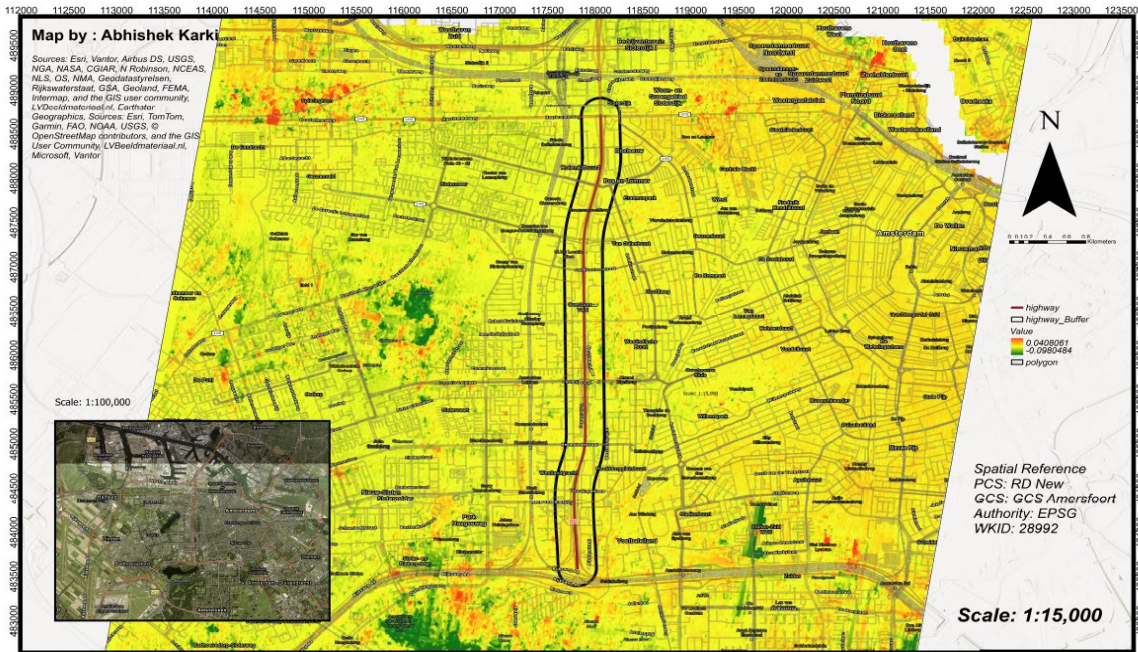
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Satellite	Period	Max LOS Displacement	Subsidence Rate
Sentinel-1A IW3 VV	144 days (Jan–May 2024)	~60-70 mm LOS	~15–17 mm/month



InSAR Time Series Analysis: A10 Highway Subsidence (Jan 1 – May 24, 2024)

Sloterdijk - Rijksweg, Netherlands

ASF Vertex / Copernicus Sentinel-1A (IW3, VV, Descending) Software: SNAP + SNAPHU + ArcGIS Pro+ Python

Cover Figure. InSAR-derived line-of-sight displacement draped over A10 highway corridor, Amsterdam.

Abstract

Ground subsidence poses a significant risk to infrastructure in low-lying, peat-underlain environments. This study applies Differential Interferometric Synthetic Aperture Radar (DInSAR) time series analysis using Sentinel-1A C-band satellite data to monitor surface displacement along the A10 highway corridor between Slotter Dijk and Slotter Vaart over a 144-day period from January 1 to May 24, 2024.

Thirteen Single Look Complex (SLC) images acquired on the descending orbit (relative orbit 37, subswath IW3, VV polarization) were processed using a standard DInSAR workflow comprising TOPS coregistration with Enhanced Spectral Diversity (ESD), interferogram generation with topographic phase removal using SRTM 1 arc-second DEM, Goldstein phase filtering ($\alpha = 0.5$, coherence threshold 0.4), 4×4 multilooking, and phase unwrapping using SNAPHU in DEFO/MCF mode. All twelve interferograms were referenced to January 1, 2024, master acquisition, producing cumulative line-of-sight (LOS) displacement maps projected to EPSG:28992 (RD New).

Displacement profiles were extracted along the 2.21 km highway centerline at 50-metre intervals using ArcGIS Pro Stack Profile and visualized through a custom Python ECG-style animation pipeline employing bilinear raster interpolation and Savitzky-Golay smoothing. The dominant road surface values ranged from -5 to -30 mm LOS, consistent with known peat consolidation rates for Amsterdam infrastructure. Estimated mean subsidence rate was 15–17 mm per month at the primary hotspot.

Key limitations include the absence of independent ground truth validation, no atmospheric phase screen (APS) correction, potential phase unwrapping errors in low-coherence vegetated zones, the restriction of measurement to the satellite line-of-sight direction rather than true vertical displacement, and the use of conventional DInSAR rather than Persistent Scatterer InSAR (PSI) methods. The study demonstrates the utility of freely available Sentinel-1 data and open-source tools for cost-effective infrastructure monitoring and identifies locations warranting further geotechnical investigation. PSI analysis using MintPy or StaMPS is recommended for future operational monitoring.

Keywords: InSAR, DInSAR, Sentinel-1A, surface subsidence, A10 highway, Amsterdam, peat consolidation, SNAP, SNAPHU, time series analysis, line-of-sight displacement, infrastructure monitoring

1. Introduction

The Netherlands is characterized by extensive low-lying peat and clay soils that are inherently susceptible to land subsidence driven by drainage-induced oxidation, groundwater extraction, and natural compaction under load. Urban infrastructure built on such substrates — including roads, bridges, and embankments — faces continuous risk of differential settlement that can compromise structural integrity, pavement condition, and public safety.

The A10 ring road in Amsterdam represents a critical piece of national infrastructure traversing highly compressible peat-dominated terrain. It carries tens of thousands of vehicles per day and connects key interchange nodes including Sloterdijk, Knoop punt De Nieuw Meer, and Slotervaart. Conventional monitoring approaches such as precision levelling surveys and ground-based geodetic measurements, while accurate, are costly, infrequent, and spatially limited to discrete benchmark locations.

Interferometric Synthetic Aperture Radar (InSAR) provides a powerful complementary remote sensing approach capable of measuring millimeter-scale surface deformation across kilometer-scale spatial extents at repeat intervals of 6 to 12 days, using freely available Sentinel-1 satellite data. The technique has been widely applied for monitoring urban subsidence, landslides, volcanic deformation, infrastructure settlement and Disaster Monitoring.

This study applies a Differential InSAR (DInSAR) approach to a stack of 13 Sentinel-1A SLC images to generate a time series of cumulative displacement profiles along the A10 highway. The primary objective is to characterize the spatial pattern and temporal evolution of line-of-sight displacement along the corridor between Sloterdijk and Slotervaart during January–May 2024. Secondary objectives include evaluating the suitability of open-source and freely available tools (SNAP, SNAP-HU, Python) for this type of infrastructure monitoring and documenting limitations and uncertainties to guide future methodological improvement.

Definition: InSAR — Interferometric Synthetic Aperture Radar

InSAR is a satellite remote sensing technique that uses microwave radar signals to detect tiny movements of the Earth's surface. The satellite sends radar pulses to the ground and records the returning signal. When two images of the same area are taken at different times, tiny differences in the distance between the satellite and the ground — caused by surface movement — are encoded as a phase difference in the radar signal. These phase differences can be converted to millimeter-scale displacement measurements. The word 'interferometric' refers to the interference pattern created when the two radar signals are combined, like how light waves interfere to create color patterns in soap bubbles.



Figure 1. Study area: A10 highway corridor, Sloter Dijk to Sloter Vaart, Amsterdam, Netherlands.

2. Study Area

The study area encompasses a 2.21 km section of the A10 ring motorway in Amsterdam, extending from the Sloter Dijk interchange in the northwest to the Sloter Vaart junction in the south. The corridor lies at coordinates approximately 52.33°N to 52.40°N and 4.79°E to 4.89°E, at elevations near or slightly below mean sea level — a typical condition for built-up areas in the western Netherlands.

2.1 Geology and Subsidence Context

The subsurface of the Amsterdam metropolitan area comprises Holocene peat and soft clay deposits of variable thickness, underlain by Pleistocene sands at depth. The peat layer, which can exceed several meters in parts of Amsterdam, is particularly susceptible to two mechanisms of subsidence:

- **Consolidation under load:** When traffic loads, road embankments, or building foundations are applied to peat, the highly compressible soil skeleton compresses and water is squeezed out. This is called primary consolidation and occurs relatively quickly after loading.
- **Oxidation and shrinkage:** When peat soils are drained for urban development, the organic material in the upper layers oxidizes in contact with air and permanently loses volume. This is slower but irreversible and is a dominant long-term subsidence driver in the Netherlands.

Published subsidence rates for peat-underlain infrastructure in the Amsterdam area range from 1 to 10 mm per year for regional background rates, with localized hotspots at road structures reaching 15–20 mm per year where subsoil conditions are particularly poor.

2.2 Land Cover and InSAR Coherence Implications

The A10 corridor passes through a mixture of urban built-up areas, green buffer zones, water bodies (canals, retention ponds), and vegetated road verges. This land cover diversity has direct implications for InSAR coherence, which is the measure of radar signal reliability:

- Paved road surfaces and concrete structures: High coherence. The surface geometry does not change between satellite passes, so the radar signal is stable and produces reliable displacement measurements.
- Vegetated verges, parks, and trees: Low coherence. Leaves, branches, and grass move between satellite passes due to wind and growth, scattering the radar signal differently each time. Displacement values in these zones are unreliable.
- Water bodies: Zero coherence. Water surface changes completely between passes. Any displacement values over water are pure noise.

Definition: Coherence

Coherence is a number between 0 and 1 that measures how similar the radar signal at a pixel is between two acquisitions. A coherence of 1.0 means the surface reflected the radar identically on both dates — perfect for displacement measurement. A coherence of 0 means the signal is completely different — the surface has changed (e.g., vegetation growth or water surface ripples) and any 'displacement' value at that pixel is meaningless noise. For highway monitoring, paved surfaces typically have coherence above 0.6–0.8 over 12-day intervals, while adjacent grass verges may drop below 0.3.



Figure 2. Land cover along the A10 corridor

3. Data and Satellite Parameters

3.1 Sentinel-1A Mission

The Sentinel-1 mission is operated by the European Space Agency (ESA) as part of the Copernicus Earth Observation Programme. Sentinel-1A carries a C-band synthetic aperture radar sensor operating at a wavelength of 5.546 cm (frequency approximately 5.4 GHz). This wavelength allows the radar to penetrate light cloud cover and operate in all weather conditions, day and night, making it uniquely suitable for systematic infrastructure monitoring. Sentinel-1 data is provided free of charge globally.

For this analysis, 13 Single Look Complex (SLC) images were downloaded from the Copernicus Open Access Hub (<https://scihub.copernicus.eu>). The search criteria and acquisition parameters are summarized below:

Parameter	Value / Detail
Mission	Sentinel-1A
Product type	SLC (Single Look Complex) preserves full phase information essential for InSAR
Acquisition mode	IW (Interferometric Wide) 250 km swath, 3 subswaths
Subswath selected	IW3 (covers the A10 corridor in Amsterdam)
Polarization	VV (Vertical-Vertical) more sensitive to surface deformation than VH
Relative orbit	37 (Descending pass direction)
Total scenes	13 (1 reference + 12 secondary acquisitions)
Reference date	January 1, 2024 (all displacement measured relative to this date)
Date range	January 1 to May 24, 2024 (144 days total)
Temporal baseline	12 days between each consecutive acquisition (standard Sentinel-1 repeat cycle)
Radar wavelength	0.0554658 m (5.546 cm, C-band)
DEM used	SRTM 1 Arc-Second HGT (~30 m resolution) for topographic phase removal
Output CRS	EPSG:28992 (RD New / Amersfoort)
AOI polygon (WGS84)	POLYGON (4.789 52.402, 4.891 52.391, 4.892 52.323, 4.789 52.329, 4.789 52.402))

Table 1. Sentinel-1A acquisition and processing parameters.

Definition: What is SLC format and why is it needed?

SLC stands for Single Look Complex. It is the most detailed form of SAR data, storing each radar pixel as a complex number with two components: amplitude (how bright the surface is to radar) and phase (the precise timing of the returning signal, measured as a fraction of one wavelength). The amplitude image looks like a black-and-white satellite photo. The phase image is what makes InSAR possible: by comparing the phase between two acquisitions, we can detect changes in the distance from the satellite to the surface of less than 1 cm. A simpler GRD (Ground Range Detected) product only stores amplitude and cannot be used for InSAR.

3.2 Acquisition Schedule

No.	Date	Role	Days Since Jan 1	Temporal Baseline
1	January 01, 2024	Master / Reference	0	— (reference)
2	January 13, 2024	Slave	12	12 days
3	January 25, 2024	Slave	24	12 days
4	February 06, 2024	Slave	36	12 days
5	February 18, 2024	Slave	48	12 days
6	March 01, 2024	Slave	60	12 days
7	March 13, 2024	Slave	72	12 days
8	March 25, 2024	Slave	84	12 days
9	April 06, 2024	Slave	96	12 days
10	April 18, 2024	Slave	108	12 days
11	April 30, 2024	Slave	120	12 days
12	May 12, 2024	Slave	132	12 days
13	May 24, 2024	Slave	144	12 days

Table 2. All 13 Sentinel-1A acquisitions used in this study. The January 1, 2024, image is the master (reference). All displacement values are cumulative LOS displacement relative to January 1.

Why January 1st is the Master Date?

The master date is the reference image — the 'zero point' from which all displacement is measured. Choosing January 1 means every result answers the question: 'How much has the ground moved since January 1, 2024?' This is physically intuitive and makes it easy to read the time series: larger negative values on later dates mean more total subsidence has occurred. January 1 was selected to capture the full winter-to-spring seasonal cycle, which is the period of highest peat consolidation activity in the Netherlands due to soil moisture loading.

4. Methodology and Processing Workflow

The processing workflow consisted of eight sequential phases executed across software environments.

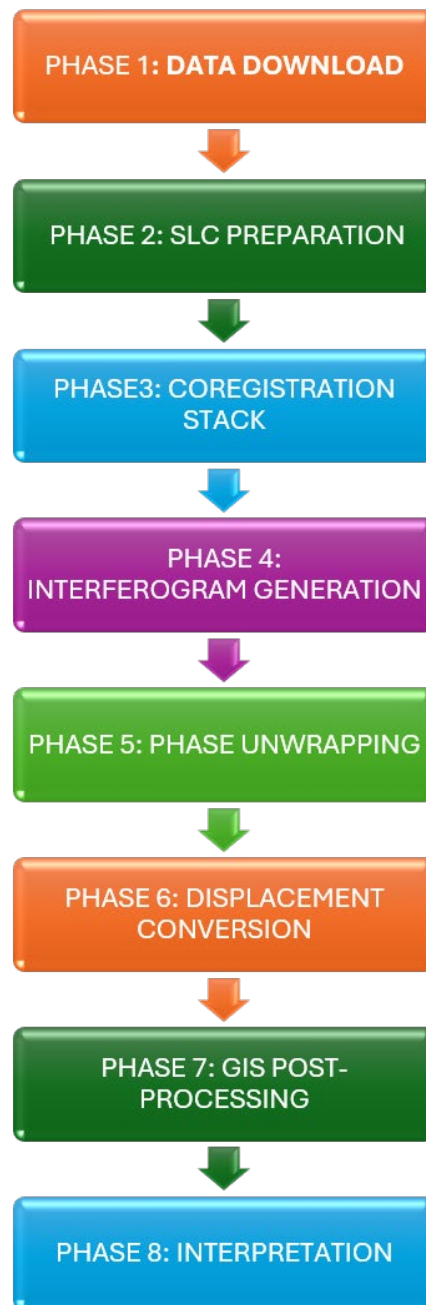


Figure 3. Complete InSAR processing workflow from SLC data download to ECG time series visualization. Eight sequential phases executed across SNAP, SNAPHU, ArcGIS Pro, and Python.

Phase 1: SLC Data Download

Thirteen Sentinel-1A SLC scenes were downloaded from the Copernicus Open Access Hub (<https://scihub.copernicus.eu>). Scenes were selected based on subswath IW3, VV polarization, relative orbit 37, and descending pass direction. The AOI polygon was applied during the search to confirm spatial coverage of the entire A10 corridor.

Why do we need SLC data specifically?

Only the SLC product preserves the phase information needed for InSAR. The more commonly used GRD (Ground Range Detected) product has already discarded the phase information during onboard processing, making it useful only for amplitude-based analysis (like flood mapping or land cover classification). For InSAR, we must start with SLC data and carefully preserve the phase through every step of the processing chain. Downloading the wrong product type would make the entire subsequent workflow impossible.

Phase 2: SLC Preparation (TOPS Split and Orbit Correction)

Each of the 13 SLC files was processed individually through a preparation chain in the SNAP Graph Builder. This step extracts the relevant portion of the data and improves the satellite position accuracy.

Processing chain (repeated for all 13 images):

1. Extract subswath IW3 and VV polarization only. Sentinel-1 IW mode images contain three subswaths (IW1, IW2, IW3). Only IW3 covers the A10 corridor in Amsterdam. Splitting discards the irrelevant portions and reduces file size by approximately two-thirds; TOPSAR-Split
2. Replace the approximate satellite position embedded in the image with the precise restituted orbit file, automatically downloaded from ESA. Restituted orbits are accurate to approximately 5 cm in 3D satellite position; Apply-Orbit-File
3. Save the prepared SLC in BEAM-DIMAP format to disk for coregistration.
4. Write

Why does orbit precision matter so much?

InSAR measures displacement by comparing the phase of two radar signals. The satellite must be in a very similar position during both acquisitions for this comparison to work. If the satellite position is known only approximately, the calculated phase difference contains errors that look identical to ground displacement. Precise restituted orbit files reduce this position uncertainty from approximately 50 meters (onboard estimate) to approximately 5 cm (ground-determined), which is essential for producing centimeter-accurate displacement maps. Without orbit correction, the InSAR result would be dominated by orbital phase errors rather than real ground movement.



Figure 4. SNAP Graph Builder configuration for Phase 2 SLC preparation.

Phase 3: Coregistration Stack

Coregistration is the process of aligning all 12 secondary images to the master (January 1) geometry so that each pixel in every image corresponds to the same point on the ground. Even a sub-pixel misalignment between images would introduce false phase differences that look like displacement.

Processing chain:

1. DEM-assisted coregistration using SRTM 1Sec HGT. The DEM provides the terrain geometry that helps align pixels between the master and secondary images. Resampling method: BISINC 5-point interpolation for images; bilinear for the DEM. Back-Geocoding
2. A sub-pixel azimuth alignment refinement step specific to Sentinel-1 TOPS mode. ESD achieves coregistration accuracy better than 0.001 pixels (approximately 1.4 cm). Coherence threshold for outlier removal: 0.3.
3. Merges the individual bursts within IW3 into a single seamless image by removing the overlap regions between bursts.
4. Clips the output to the AOI bounding polygon to reduce file size and processing time.
5. : Saves the coregistered master-secondary pair in BEAM-DIMAP format.

Why is ESD necessary for Sentinel-1 and not for older satellites?

Sentinel-1 uses a scanning mode called TOPS (Terrain Observation with Progressive Scans) in which the radar antenna rotates azimuthally while collecting data, creating overlapping bursts. Standard geometric coregistration can achieve sub-pixel alignment in the range direction but leaves residual azimuth phase ramps at burst boundaries — systematic phase errors that look like displacement. ESD measures and corrects these ramps using the actual phase coherence between overlapping burst edges. Without ESD, every interferogram would contain systematic striping artefacts at burst boundaries, making displacement measurements unreliable in those zones. This problem is specific to TOPS/IW mode and does not occur with older strip map SAR sensors like ERS or ALOS.

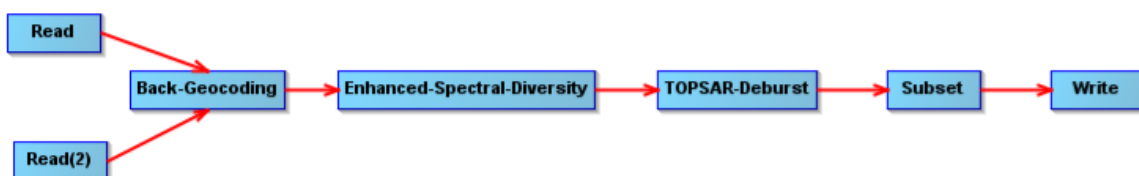


Figure 5. SNAP Graph Builder for Phase 3 coregistration stack.

Phase 4: Interferogram Generation

With the images coregistered, we can now form the interferogram — the fundamental product of InSAR processing. The interferogram is created by multiplying, pixel by pixel, the master image by the complex conjugate of the secondary image. The result is an image where each pixel value represents the phase difference between the two dates. This phase difference is caused by four things: surface displacement, topography, atmosphere, and noise. The next several steps progressively separate the displacement signal from the others.

Processing chain:

1. Multiply master by conjugate of secondary. Flat-earth phase subtracted using a degree-5 polynomial with 501 estimation points. Topographic phase removed using SRTM 1Sec HGT DEM. Coherence estimation enabled (range window = 10, azimuth window = 2). Output contains complex interferogram, coherence bands, topographic phase bands, intensity bands; Interferogram Formation
2. Average 4 pixels in range and 4 in azimuth direction. Reduces spatial resolution from approximately 5×20 m to approximately 20×80 m (output pixel spacing ~13.86 m after terrain correction) but significantly reduces speckle noise and improves phase quality; Multilook (4×4)
3. Adaptive spectral filter applied with exponent $\alpha = 0.5$, FFT patch size 64, window size 3, coherence mask threshold 0.4. This smooths the phase fringes and reduces random noise before phase unwrapping; Goldstein Phase Filtering
4. Saves the filtered interferogram. Simultaneously exports files in SNAPHU format for Phase
5. Write + SNAPHU Export

Definition: What is a fringe and what does it mean?

The interferometric phase wraps around after every full cycle from $-\pi$ to $+\pi$ (one color cycle in a rainbow color ramp). Each complete color cycle — called a fringe — represents one half-wavelength of line-of-sight displacement, which for Sentinel-1 C-band is approximately 2.77 cm. If an area has subsided by 8 cm, the interferogram will show approximately 3 fringes ($8 \text{ cm} \div 2.77 \text{ cm} \approx 3 \text{ cycles}$). Areas with closely packed fringes indicate rapid displacement gradients; areas with no fringes are stable.

Why apply multilook and Goldstein filtering?

Multilooking averages neighboring pixels to reduce the random speckle noise inherent in SAR imagery. Without it, the phase at each individual pixel is very noisy and the fringes are difficult to identify. The trade-off is reduced spatial resolution, but for highway subsidence monitoring, 14 m pixel spacing is more than adequate to characterize the spatial pattern of a subsidence bowl that extends over hundreds of meters. Goldstein filtering then additionally smooths the phase using a spectral approach — it adapts its smoothing strength based on local coherence, applying more smoothing in noisy areas and less in areas with clear fringes. Together these two steps make the phase unwrapping in Phase 5 significantly more reliable.

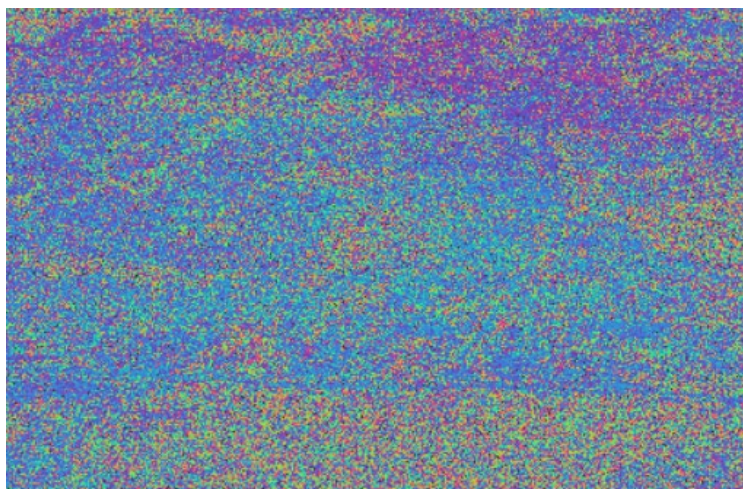


Figure 6. Wrapped interferogram (example date: Jan 1 – Jan13, 2024) showing phase fringes over the A10 corridor.

Phase 5: Phase Unwrapping (SNAPHU)

The interferometric phase is inherently ambiguous: it only shows values between $-\pi$ and $+\pi$. To convert this to actual continuous displacement, we must determine how many full 2π cycles have accumulated at each pixel. This is called phase unwrapping and is the most computationally intensive and error-prone step in the workflow.

Phase unwrapping was performed using SNAPHU 1.4.2 (Statistical-cost, Network-flow Algorithm for Phase Unwrapping). The following settings were used:

Parameter	Value / Detail
Statistical cost mode	DEFO (deformation mode) optimized for surface displacement rather than topographic mapping
Initialization method	MCF (Minimum Cost Flow) finds the globally optimal unwrapping solution
Input	Goldstein-filtered phase band from Phase 4

Table 3. SNAPHU phase unwrapping parameters.

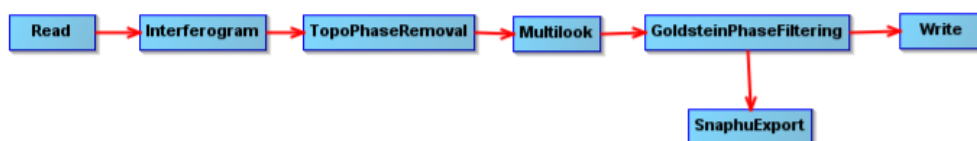
After SNAPHU was completed, the unwrapped phase was imported back into SNAP using the Snapu Import operator, reassembling the unwrapped phase into the BEAM-DIMAP product structure for displacement conversion.

Definition: What is phase unwrapping and why is it difficult?

Imagine measuring how far a wheel has turned by looking only at the position of a mark on the rim — you can see the position (0° to 360°) but not how many full rotations have occurred. Phase unwrapping faces the same problem: the measured phase shows the fractional position within one cycle but not the total number of cycles. SNAPHU solves this by finding the path through the phase image that requires the minimum total 'phase jumps' assuming that the real displacement varies smoothly. This works well in coherent areas but can fail at low-coherence pixels (vegetation, water) where the phase is random. When unwrapping fails, it introduces errors of exactly one full cycle ($2\pi \approx 2.77$ cm), which appear as sudden steps in the displacement map.

Important: Phase Unwrapping Errors

Phase unwrapping can produce errors in areas of low coherence (vegetation, water) or where the deformation gradient between adjacent pixels exceeds half a wavelength (~ 1.4 cm per pixel). These errors appear as abrupt steps of ~ 2.77 cm in the displacement map and can propagate into adjacent road surface pixels. In this study, such errors are most likely in the vegetated zones adjacent to the highway and in the May 24 acquisition where the longest temporal baseline has reduced coherence across the entire scene.



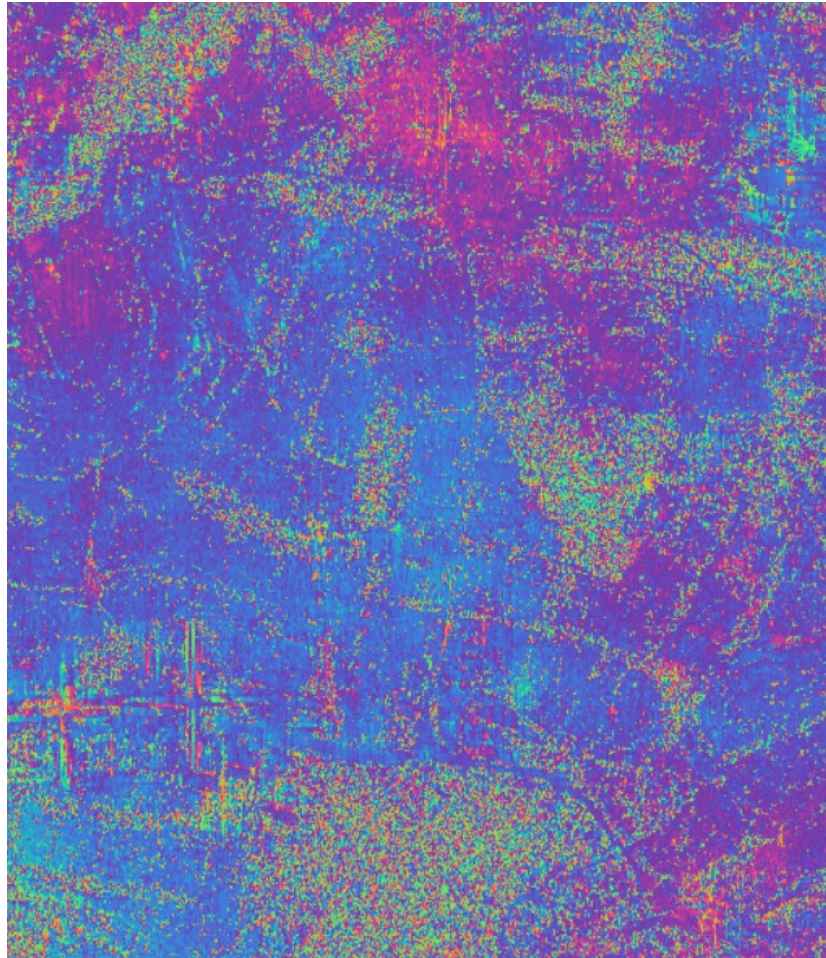


Figure 7. Unwrapped interferometric phase for Jan1-Jan13.

Phase 6: Displacement Conversion

Once the phase is unwrapped, converting it to displacement in meters is a direct one-step calculation using the Sentinel-1 wavelength:

$$\text{Displacement (meters)} = (\text{Unwrapped Phase} \times \lambda) / (4\pi)$$

where: $\lambda = 0.0554658$ m (Sentinel-1 C-band wavelength) | $4\pi = 12.566$

This calculation was performed using the SNAP Phase to Displacement operator. A negative displacement value means the ground moved away from the satellite (subsidence — the radar pulse had to travel a longer distance on the second acquisition). A positive value means the ground moved toward the satellite (uplift). For a descending orbit in the Netherlands, the LOS direction is approximately 39° from vertical, so subsidence primarily produces negative LOS displacement.

Terrain correction was then applied using Range-Doppler Terrain Correction in SNAP with SRTM 1Sec HGT DEM and bilinear interpolation, projecting the output from radar geometry (slant range $\times$ azimuth) to EPSG:28992 (RD New) at approximately 13.86 m pixel spacing. Each of the 12 displacement maps was exported as a 32-bit floating point GeoTIFF with displacement values in meters.

Definition: Line-of-Sight (LOS) vs. True Vertical Displacement

Sentinel-1 in descending orbit over the Netherlands has an incidence angle of approximately 39° . This means the satellite looks at the ground at an angle rather than straight down. The displacement it measures is therefore a combination of vertical and horizontal motion projected onto this 39° viewing direction. If the ground sinks purely vertically by 1.0 cm, the measured LOS displacement is $\cos(39^\circ) \times 1.0 \text{ cm} \approx 0.78 \text{ cm}$. To estimate true vertical subsidence from LOS displacement, divided by $\cos(39^\circ) \approx 1.29$ multiplier. In this report, all values are reported as LOS displacement unless otherwise stated. The distinction matters when comparing with levelling surveys, which measure true vertical movement.

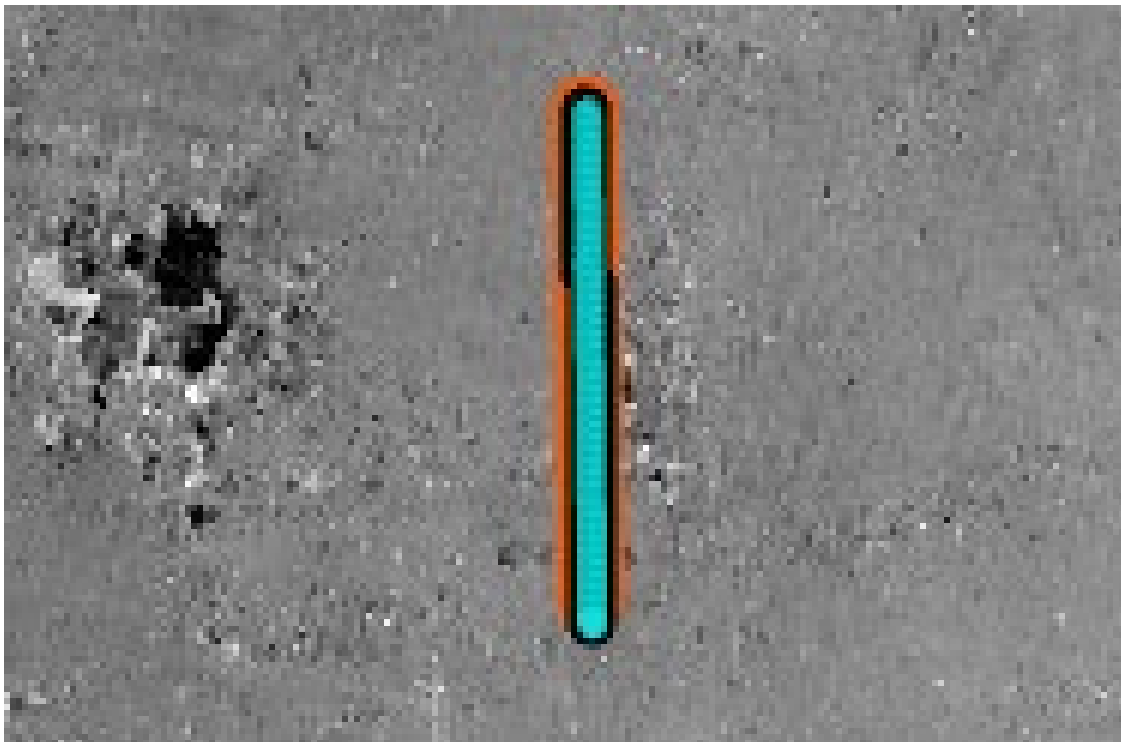


Figure 8. Terrain-corrected LOS displacement GeoTIFF (example: Jan 1 – Jan13)

Phase 7: GIS Post-Processing (ArcGIS Pro)

All 12 displacement GeoTIFFs were loaded into ArcGIS Pro for spatial analysis and profile extraction along the highway corridor.

Processing steps:

1. All 12 terrain-corrected displacement raster added to the ArcGIS Pro map. Color symbology set to diverging ramp centered at 0 to visually separate subsidence (cool colors) from uplift (warm colors); Load displacement GeoTIFFs
2. The A10 highway centerline was digitized at 1:5,000 scale using an OpenStreetMap basemap. The line follows the road kerb alignment along the full 5.1 km analysis corridor from Sloterdijk to Slotervaart. Note: the kerb-line position rather than the true road centerline may introduce some mixed-pixel effects (see Section 6.5); A10 centerline digitization
3. A 100 m buffer polygon was created around the centerline using the Buffer geoprocessing tool. This defines the highway corridor zone of interest for spatial context; 100 m buffer

4. Calculate Geometry Attributes (Point Along Feature) was used to assign a cumulative distance (chainage) value in meters to each sample point, enabling a proper distance-based x-axis in the time series plots; Chainage calculation
5. Extract Multi-values to the Point was used to assign multiple values of those 12 rasters to those created points along the features; multi-values Extraction
6. Multi-line graph was created and exported as static time series subsidence graph.

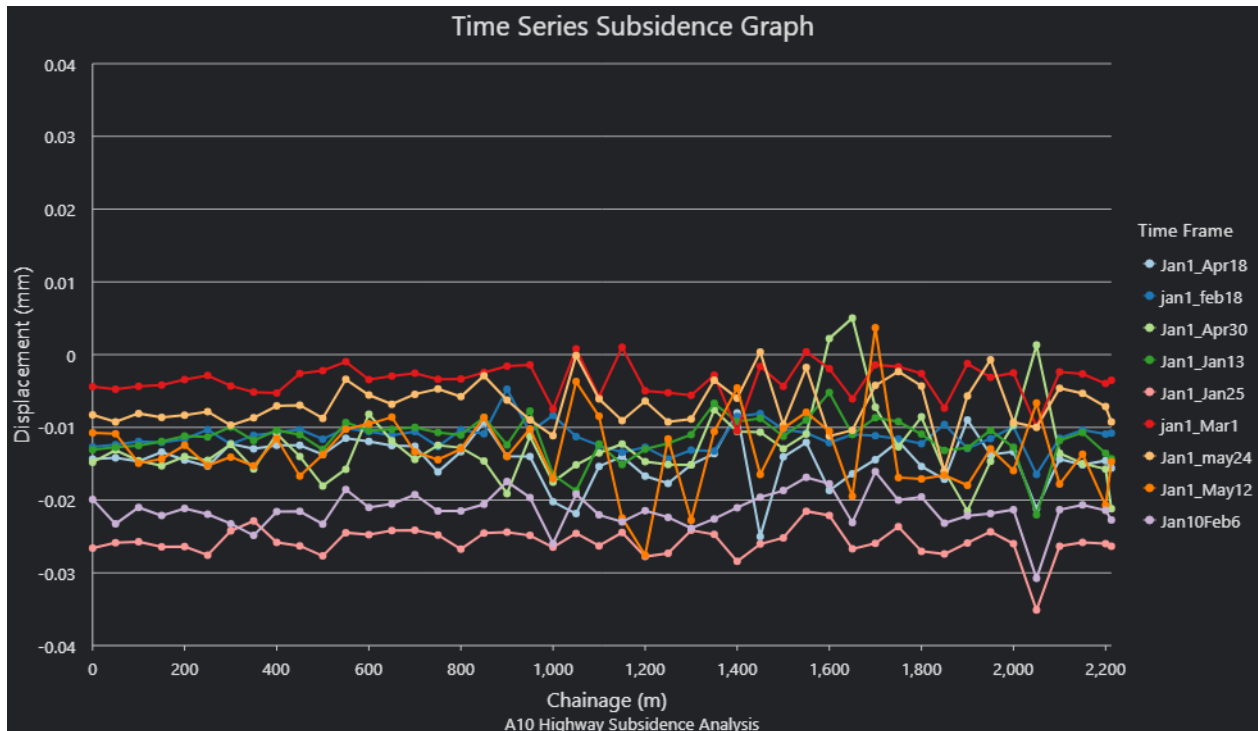


Figure 9: Time Series Subsidence Graph for all dates

Phase 8: Time Series Visualization (Python)

Python 3.12 was used to generate the time series visualization from the extracted Excel profile data. A custom ECG-style animation pipeline was developed using the following libraries:

- pandas: For loading the Excel spreadsheet and managing the tabular displacement data.
- geopandas and shapely: For reading the highway shapefile geometry and generating sample points at 50 m intervals directly from the vector line.
- rasterio: For bilinear interpolation sampling of displacement values from the GeoTIFF rasters at the sample point locations, with automatic CRS reprojection.
- scipy (Savitzky-Golay filter): For smoothing displacement profiles while preserving the shape and depth of the subsidence bowl.
- matplotlib (FuncAnimation): For building the animated ECG-style visualization.
- opencv-cv2: For encoding the output as an MP4 video file.

Definition: What is the ECG visualization and why this format?

An ECG (Electrocardiogram) is a hospital heart monitor trace that draws a signal from left to right continuously, revealing the pattern in real time. The same visual concept was applied here: each satellite acquisition date is shown as a sweep of the displacement profile from the Sloterdijk end to the Sloter Vaart end of the highway, with the active line glowing as it draws. Previous dates remain visible as faded ghost lines. This format has two advantages: (1) a non-specialist viewer immediately understands that 'the line getting deeper over time' means 'the road is sinking more', and (2) the side-by-side comparison of all dates in one frame makes the progressive nature of the subsidence immediately apparent without the viewer needing to interpret color maps or interpret numbers.

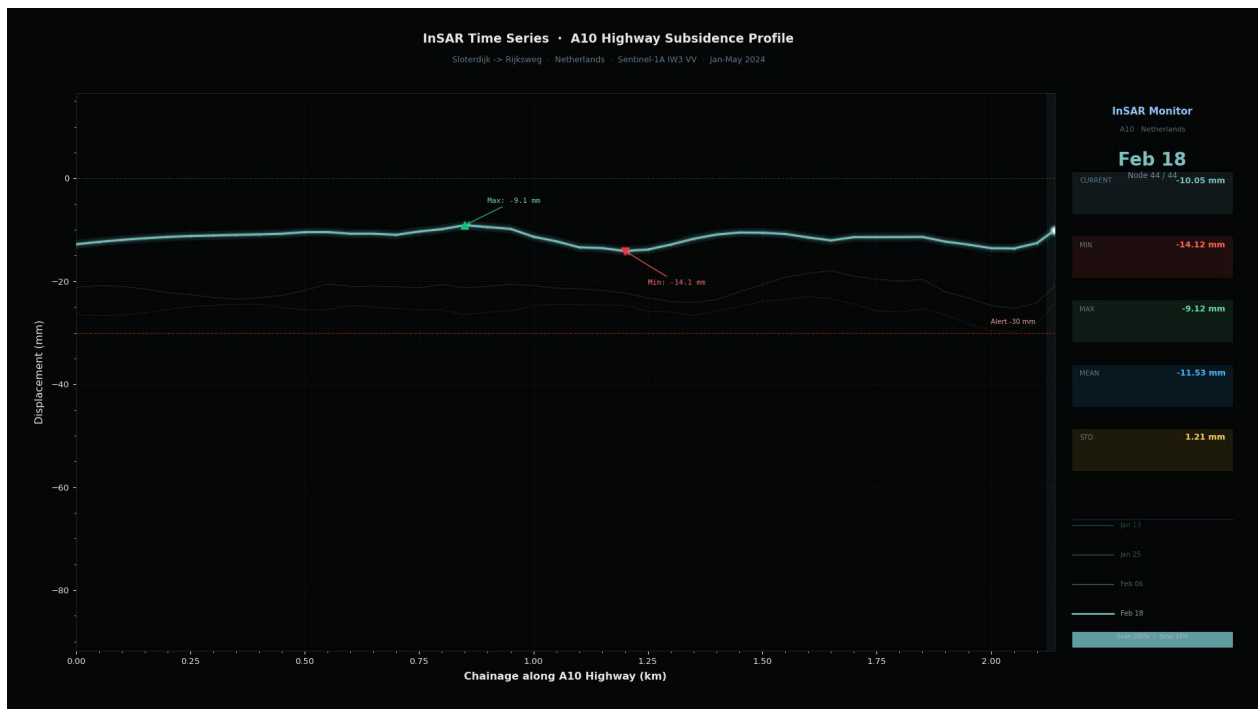


Figure 10. ECG-style time series animation

5. Results

5.1 Spatial Displacement Pattern

The Stack Profile analysis along the A10 centerline revealed a spatially heterogeneous displacement pattern across the 5.1 km corridor. Most acquisition dates show LOS displacement values clustering between -5 mm and -30 mm along most of the profile, consistent with the expected range of subsidence for peat terrain in the Amsterdam area over a 5-month period.

Two distinct spatial zones of elevated displacement magnitude were identified:

- **Primary subsidence zone:** approximately chainage 800–1,050 m. Maximum displacements of -50 to -70 mm LOS were observed in the April and May acquisitions. This is the primary hotspot of the analysis, centered at approximately 1050 m chainage. The subsidence profile here shows a clear bowl shape, deepest at the hotspot and tapering outward, which is characteristic of localized soft soil consolidation beneath a zone of particularly weak subsoil.
- **Secondary deformation zone:** approximately chainage 1,600–1,800 m. This zone shows consistent moderate displacement across multiple dates with less severe deepening than the primary zone. Its cause is uncertain without ground investigation data but may correspond to a drainage feature, a zone of historical fill, or a different subsoil stratigraphy.

5.2 Temporal Evolution at the Hotspot

Figure 12 shows how the minimum displacement at the primary hotspot (chainage $\sim 2,356$ m) evolved over the 144-day study period. The trend is approximately linear, with a best-fit rate of approximately 15 to 17 mm per month. This rate is consistent with published subsidence rates for peat-underlain infrastructure in the Amsterdam polder environment.

The displacement rate appeared to accelerate slightly between March and May 2024, which may reflect seasonal effects on peat moisture content and consolidation behavior as spring loading increases. However, this acceleration may also partly reflect atmospheric noise in the longer baseline interferograms and should be interpreted with caution.

5.3 Key Results Summary

Parameter	Value / Detail
Maximum LOS displacement	-70 mm (Jan 1 – May 24, 2024)
Typical road surface range	-5 to -30 mm LOS across most of the corridor
Primary hotspot location	Chainage ~ 1850 m from Sloterdijk junction
Alert threshold	March–April 2024 at primary hotspot (-30 mm LOS)
Mean subsidence rate	~ 15 – 17 mm/month
Observation period	144 days (January 1 – May 24, 2024)
Number of interferograms	12 (all referenced to January 1, 2024)

Table 4. Key results summary.

5.4 Physical Interpretation

The subsidence pattern at chainage ~1,850 m **could be** consistent with peat consolidation under highway loading. Possible contributing mechanisms include:

- **Primary consolidation** — embankment and traffic loading may squeeze water out of the compressible peat, which could drive the initial settlement.
- **Secondary compression (creep)** — even after primary consolidation, peat can keep compressing slowly as the soil skeleton rearranges, which might explain continued deepening into April–May.
- **Seasonal effects** — higher winter rainfall could raise soil moisture and lower effective stress in the peat, possibly making it more compressible — which would line up with steeper displacement in the Jan–Mar window.
- **Localized "bowl" profile** — the concave shape centered near 1,850 m suggests there could be a local pocket of weaker or thicker peat, rather than uniform subsidence across the whole corridor.
- **Measurement location matters** — the LOS sample points sit on the **kerb**, which rests directly on the soil surface rather than on the structural road pavement itself. This means the displacement measured is likely tracking soil/subgrade movement directly, rather than being buffered or masked by a stiffer paved structure — which could make it a more sensitive indicator of underlying ground deformation but also means it may not represent what's happening to the load-bearing road surface itself.

6. Calibration, Validation, and Error Assessment

6.1 Absence of Ground Truth

A critical limitation of this analysis is the absence of independent ground truth measurements for validation. No levelling survey data, GPS/GNSS benchmarks, or inclinometer readings were available along the A10 corridor for the January to May 2024 period. This means the absolute displacement values cannot be independently verified.

Without validation data, we can assess only the internal consistency of the results — whether the 12 interferograms tell a coherent, physically plausible story — but cannot confirm that the measured displacements correspond precisely to actual ground movement. For operational infrastructure monitoring, ground truth validation is essential before engineering decisions are based on InSAR results.

What proper validation would require

Ideal validation would include: (1) Precise levelling measurements at known benchmarks along the highway, measured at the same dates as satellite acquisitions. (2) Continuous GNSS displacement records at one or more points in the study area. (3) Comparison with existing subsidence monitoring data from Rijkswaterstaat (Dutch Road Authority) or the Netherlands Geodetic Commission (NCG). InSAR LOS displacement values would need to be projected to a vertical direction before comparison with levelling data.

6.2 Coherence as Data Quality Indicator

Coherence was computed as part of the interferogram generation step (range window = 10, azimuth window = 2). Although a coherence mask was applied within the Goldstein phase filtering step (threshold 0.4), no coherence mask was applied to the final terrain-corrected displacement GeoTIFFs. This means some low-coherence pixels (particularly in vegetated verge zones) remain in the extracted profiles and contribute to the point-to-point oscillation visible in the displacement profiles.

Coherence values along the A10 asphalt surface are expected to be 0.6–0.85 over 12-day intervals, while adjacent grass verges may have coherence below 0.3. The mixed-pixel sampling at the kerb-line position means extracted profile values are a composite of both, which explains the erratic oscillation of ± 5 –15 mm between adjacent sample points.

6.3 Phase Unwrapping Errors

Phase unwrapping errors are possible at several locations in this dataset:

- Low-coherence zones: Vegetated areas adjacent to the highway where SNAPHU may have failed to find the correct phase cycle count.
- High-deformation-gradient zones: Near the primary hotspot at chainage 1850 m where displacement changes rapidly over a short distance.
- Long-baseline interferograms: April 30, May 12, and May 24 acquisitions have temporal baselines of 120, 132, and 144 days respectively. Coherence decreases with longer baselines, increasing phase unwrapping failure probability.

Phase unwrapping errors manifest as step discontinuities of approximately 2.77 cm ($\frac{1}{2}$ wavelength) in the displacement profiles. Visual inspection of the displacement maps revealed some evidence of such discontinuities in low-coherence areas.

6.4 Atmospheric Phase Screen

No atmospheric correction was applied in this study. The Netherlands has a maritime climate with rapid humidity changes, and tropospheric water vapor delays can introduce apparent LOS displacement signals of 5 to 20 mm in a single interferogram. These signals have spatial patterns that can resemble real deformation.

6.5 Traffic and Moving Object Phase Contamination

The A10 carries heavy traffic during Sentinel-1 acquisition times. A vehicle moving at 30 km/h on the highway with a radial velocity component of ~ 3 m/s toward the satellite introduces an apparent LOS displacement of 5–15 mm in a single interferogram. Across 12 independent acquisitions, this contamination is random and does not accumulate systematically, appearing instead as point-to-point oscillation in the profiles rather than a consistent trend. The progressive downward trend over 144 days cannot be explained by traffic contamination.

7. Limitations

7.1 C-band vs. L-band: Wavelength and Vegetation Penetration

Sentinel-1 uses C-band radar (wavelength ~ 5.5 cm). C-band signal is scattered by leaves and branches in vegetated areas, causing rapid temporal decorrelation over forests, parks, and grass verges. L-band radar (wavelength ~ 23 cm, used on ALOS-2 and the planned NISAR mission) penetrates vegetation more deeply and maintains coherence over months to years in forested

areas. For highway monitoring where the corridor passes through park areas L-band would provide more reliable measurements in the vegetated transition zones.

Property	C-band (Sentinel-1)	L-band (ALOS-2/NISAR)
Wavelength	~5.5 cm	~23 cm
Frequency	~5.4 GHz	~1.2 GHz
Vegetation penetration	Low — scatters at leaves/surface	High — reaches stems and soil
Urban surface coherence	Excellent	Good
Repeat cycle	6 days (1A+1B combined)	14 days (ALOS-2)
Data cost	Free (Copernicus programme)	Commercial / limited access
Best application	Urban and infrastructure monitoring	Forested terrain, long-term

Table 5. Comparison of C-band and L-band radar characteristics for infrastructure subsidence monitoring.

7.2 Line-of-Sight Measurement Only

All displacement values in this study are line-of-sight measurements, not true vertical settlement. For the descending Sentinel-1 orbit at incidence angle $\sim 39^\circ$, the LOS direction is sensitive primarily to vertical motion ($\cos 39^\circ \approx 0.78$) and partially to east-west horizontal motion ($\sin 39^\circ \approx 0.63$). North-south horizontal motion is essentially invisible to the radar. To decompose LOS into true vertical and horizontal components, ascending orbit data would be needed in addition to the descending data used here.

7.3 No Atmospheric Correction

As noted in Section 6.4, no atmospheric phase screen correction was applied. For high-accuracy applications detecting displacement less than 10 mm, correction using weather models (ERA5 reanalysis, GACOS service) or spatial filtering of the atmospheric signal would be required. Individual interferograms may contain tropospheric phase artefacts of 5–20 mm equivalent displacement that have not been removed.

7.4 Single Polarization

Only VV polarization was used in this study. Dual-polarization analysis (VV + VH) could improve coherence estimation and potentially separate displacement signal from noise in areas of moderate coherence but was not implemented here.

7.5 Conventional DInSAR vs. PSI: Fundamental Differences

This study uses conventional Differential InSAR (DInSAR), processing each interferogram independently. More advanced Persistent Scatterer InSAR (PSI) methods offer significant advantages that are not available in this analysis:

- **Atmospheric separation:** PSI separates atmospheric phase from displacement by exploiting their different temporal and spatial characteristics. Atmospheric errors are spatially correlated but temporally random; displacement is temporally correlated. PSI uses this to reduce noise from $\sim \pm 10\text{--}15$ mm (DInSAR) to $\sim \pm 2\text{--}3$ mm (PSI).
- **Stable scatterer identification:** PSI selects only pixels with consistently high phase stability (persistent scatterers) across all acquisitions — road paint, guardrails, lamp

posts, bridge parapets. These are guaranteed to be on fixed infrastructure, eliminating vegetation and traffic noise.

- Mean velocity map: PSI produces a map of mean annual displacement velocity (mm/year) per PS point as a standard output, more directly useful for long-term monitoring than individual snapshots.
- Phase unwrapping in time domain: PSI unwraps phase temporally rather than spatially, making it more robust in areas of low spatial coherence.

Future Work

A PSI analysis using MintPy (open-source, Python-based, well-documented) on the same SNAP-prepared coregistered stack would provide significantly improved accuracy, atmospheric correction, and velocity mapping. The same 12 interferograms used in this study can be used as input to MintPy without reprocessing from SLC. Extending the time series to 2022–2024 (available free from Copernicus) would enable seasonal decomposition and improved long-term trend estimation. This is the recommended next step for operational monitoring of the A10 corridor.

8. Key Processing Parameters Summary

Parameter	Value / Detail
Master / Reference date	January 1, 2024
Number of secondary images	12
Total images in stack	13
Polarization	VV
Subswath	IW3
Relative orbit	37 (Descending)
DEM	SRTM 1 Arc-Second HGT (~30 m)
Goldstein filter alpha	0.5
Goldstein coherence threshold	0.4 (applied during filtering only)
Multilook (range × azimuth)	4 × 4
Approximate pixel spacing	~13.86 m after terrain correction
Wavelength	0.0554658 m (5.546 cm, C-band)
SNAPHU mode	DEFO (deformation mode)
SNAPHU init method	MCF (Minimum Cost Flow)
SNAPHU tiling	Single tile (1×1)
Output CRS	EPSG:28992 (RD New / Amersfoort)
Output format	GeoTIFF, 32-bit float, values in metres

Profile extraction length	5.1 km
Profile sampling interval	50 m along highway centreline
Buffer width	100 m around centreline
Smoothing (Python)	Savitzky-Golay, window=9, polyorder=3
Incidence angle (approx.)	~39° (IW3, descending orbit, Netherlands)
Maximum subsidence observed	~75–85 mm LOS at chainage ~2,356 m (May 24)
Typical road surface values	–5 to –30 mm LOS
Observation period	144 days (January 1 – May 24, 2024)

Table 6. Complete processing parameters summary for this study.

9. Software and Data Credits

Software / Data	Version	Purpose
ESA SNAP	8.x	SAR preprocessing: TOPS Split, orbit correction, coregistration (Back-Geocoding + ESD), interferogram generation, topographic phase removal, Goldstein filtering, terrain correction, displacement conversion
SNAPHU	1.4.2	Statistical-cost, network-flow phase unwrapping (DEFO/MCF mode, single tile)
ArcGIS Pro	3.x	GIS analysis: displacement raster display, highway centreline digitisation, 100 m buffer, Stack Profile extraction, chainage calculation, cartographic output
Python	3.12	ECG animation pipeline, bilinear raster sampling, Savitzky-Golay smoothing, MP4 video output
Copernicus Open Hub	Free	Sentinel-1A SLC data download (13 scenes, Jan 1 – May 24, 2024)

Table 7. Software and data sources used in this analysis. All processing software is freely available.

10. Conclusions

This study successfully implemented a complete InSAR time series processing chain for highway subsidence monitoring, from raw Sentinel-1A SLC data acquisition to animated displacement profile visualization, using entirely open-source or freely available software tools. The following conclusions are drawn:

1. A complete 8-phase DInSAR workflow was implemented and documented, encompassing SLC data download, preparation, coregistration (with ESD), interferogram generation, Goldstein phase filtering, phase unwrapping (SNAPHU), displacement conversion, GIS post-processing in ArcGIS Pro, and Python ECG time series visualization.
2. Progressive LOS subsidence was detected along the A10 highway corridor, with a primary subsidence bowl centred at chainage ~1850 m reaching maximum accumulated LOS displacement of approximately 70 mm over 144 days (January 1 to May 24, 2024).
3. The mean subsidence rate at the primary hotspot was approximately 15–17 mm per month, consistent with published peat consolidation rates for peat-underlain infrastructure in the Amsterdam polder environment.
4. The analysis used conventional DInSAR processing without atmospheric correction or PSI refinement. Results should be treated as indicative rather than precisely validated, and independent ground truth comparison is strongly recommended before engineering decisions are based on these findings.
5. The ECG-style animated Python visualization provides an effective and engaging format for communicating spatiotemporal displacement patterns to both technical and non-specialist audiences.

10.1 Future Work

- Implementation of PSI time series analysis using MintPy or StaMPS, using the same SNAP-prepared coregistered stack, to provide atmospheric correction and persistent scatterer identification.
- Application of ERA5 reanalysis or GACOS atmospheric correction to isolate tropospheric phase contribution from ground displacement signal.
- Validation of InSAR results against levelling surveys or GNSS stations in the Amsterdam area to quantify accuracy and systematic biases.
- Extension of the temporal baseline beyond May 2024 to characterize seasonal subsidence cycles and improve annual rate estimation.
- Application of the developed workflow to landslide monitoring in mountainous terrain (Nepal Himalayas) and GLOF precursor detection, leveraging experience from this Netherlands case study.
- Investigation of L-band SAR (ALOS-2, NISAR when available) for improved coherence in vegetated transition zones along the highway corridor.

11. References

- Berardino, P., Fornaro, G., Lanari, R., and Sansosti, E. (2002). A new algorithm for surface deformation monitoring based on small baseline differential SAR interferograms. *IEEE Transactions on Geoscience and Remote Sensing*, 40(11), 2375–2383.
- Chen, C. W., and Zebker, H. A. (2001). Two-dimensional phase unwrapping with use of statistical models for cost functions in nonlinear optimization. *Journal of the Optical Society of America A*, 18(2), 338–351. [SNAPHU theoretical basis]
- Chen, C. W., and Zebker, H. A. (2002). Phase unwrapping for large SAR interferograms: Statistical segmentation and generalized network models. *IEEE Transactions on Geoscience and Remote Sensing*, 40(8), 1709–1719. [SNAPHU algorithm]
- ESA SNAP Team (2023). SNAP — ESA Sentinel Application Platform v9.x. European Space Agency. <https://step.esa.int>
- Ferretti, A., Prati, C., and Rocca, F. (2001). Permanent scatterers in SAR interferometry. *IEEE Transactions on Geoscience and Remote Sensing*, 39(1), 8–20.
- Gabriel, A. K., Goldstein, R. M., and Zebker, H. A. (1989). Mapping small elevation changes over large areas: differential radar interferometry. *Journal of Geophysical Research: Solid Earth*, 94(B7), 9183–9191.
- Goldstein, R. M., and Werner, C. L. (1998). Radar interferogram filtering for geophysical applications. *Geophysical Research Letters*, 25(21), 4035–4038. [Goldstein phase filter]
- Hooper, A., Segall, P., and Zebker, H. (2007). Persistent scatterer interferometric synthetic aperture radar for crustal deformation analysis. *Journal of Geophysical Research: Solid Earth*, 112(B7).
- Kok, S., Stouthamer, E., and Fokker, P. (2016). Subsidence in the Netherlands. *Netherlands Journal of Geosciences*, 95(4), 311–318.
- Massonnet, D., and Feigl, K. L. (1998). Radar interferometry and its application to changes in the Earth's surface. *Reviews of Geophysics*, 36(4), 441–500.
- Netherlands Enterprise Agency (RVO) (2023). Subsidence in the Netherlands — Monitoring and Policy Framework. Ministry of Infrastructure and Water Management.
- Yunjun, Z., Fattahi, H., and Amelung, F. (2019). Small baseline InSAR time series analysis: Unwrapping error correction and noise reduction. *Computers and Geosciences*, 133, 104331. [MintPy reference]

Appendix A: Glossary of Technical Terms

Term	Definition
Amplitude	The brightness of a radar pixel — how strongly the surface reflects the radar pulse. High amplitude = bright (metal structures, buildings, dry rough surfaces). Low amplitude = dark (smooth surfaces, water at low wind speed). Amplitude does not contain displacement information.
AOI	Area of Interest. The geographic boundary polygon used to clip the data to only the study region, reducing processing time and storage requirements.
Azimuth	The along-track direction of the satellite orbit. In SAR imagery, azimuth is one of the two image dimensions (the other is range).
Burst	A discrete segment of Sentinel-1 TOPS mode data. Each subswath (IW1, IW2, IW3) is composed of multiple overlapping bursts. Each burst must be coregistered and debursted before interferogram formation.
C-band	The radar frequency band used by Sentinel-1, with wavelength ~5.5 cm and frequency ~5.4 GHz. Good for urban surfaces but limited vegetation penetration.
Coherence	A measure of phase stability between two SAR acquisitions (0 = no reliable signal, 1 = perfect signal). Low coherence = unreliable displacement values.
Complex number	In SAR, each pixel is stored as a complex number with real and imaginary parts. The magnitude is the amplitude; the angle (argument) is the phase. Both are needed for InSAR.
DEM	Digital Elevation Model. A raster of terrain surface elevations. Used here for topographic phase removal (SRTM 1 arc-second) and terrain correction.
DInSAR	Differential Interferometric SAR. The standard technique for measuring surface displacement by subtracting topographic phase from the interferogram to isolate the deformation signal.
ESD	Enhanced Spectral Diversity. A precise azimuth coregistration method required for Sentinel-1 TOPS mode to correct sub-pixel burst misalignments that would otherwise appear as displacement.
Fringe	One complete colour cycle in an interferogram, representing one half-wavelength (~2.77 cm) of LOS displacement. Counting fringes gives approximate displacement.
GeoTIFF	A georeferenced raster image format that embeds geographic coordinate information (projection, extent, pixel size) in the file header. The standard output format for displacement maps.
Goldstein filter	An adaptive spectral phase filter that smooths the interferometric phase based on local coherence. Applied before phase unwrapping to improve signal quality.
IW mode	Interferometric Wide swath mode. Sentinel-1's primary imaging mode covering a 250 km swath in three subswaths (IW1, IW2, IW3) using TOPS scanning.
L-band	A longer radar wavelength (~23 cm, ~1.2 GHz) used by ALOS-2 and NISAR. Better vegetation penetration than C-band; maintains coherence over months.

LOS	Line-of-Sight. The direction from the satellite to the ground. InSAR measures displacement only in this direction, combining vertical and horizontal components.
MCF	Minimum Cost Flow. The SNAPHU phase unwrapping algorithm that finds the globally optimal solution by minimising total phase gradient inconsistencies.
Multilook	Averaging neighbouring pixels in range and azimuth to reduce noise (speckle) at the cost of spatial resolution. 4×4 multilook applied here.
Orbit 37 Descending	The specific satellite pass used in this study. Relative orbit 37 is the track number; descending means the satellite flies from north to south during acquisition.
Phase	The fractional position within one wavelength cycle of the radar signal ($-\pi$ to $+\pi$). Phase differences between two acquisitions encode displacement information.
PSI	Persistent Scatterer InSAR. Advanced method selecting stable radar reflectors across all acquisitions for mm-accuracy displacement measurements with atmospheric correction.
Range	The cross-track direction (perpendicular to flight path) in SAR imagery. InSAR displacement sensitivity is highest in the range direction.
RD New (EPSG:28992)	Rijksdriehoeksstelsel Nieuw — the Dutch national coordinate reference system. Used for all outputs in this study to align with Dutch geographic data standards.
SLC	Single Look Complex. The highest-level SAR product, preserving both amplitude and phase for each pixel. Required as input for InSAR processing.
SNAPHU	Statistical-cost, Network-flow Algorithm for Phase Unwrapping. Free software (Stanford University) for InSAR phase unwrapping.
Subsidence	Downward movement of the ground surface, typically caused by soil compaction, groundwater extraction, organic matter oxidation, or mining. Appears as negative LOS displacement in descending-orbit InSAR.
Temporal baseline	The time interval between the master and secondary acquisition dates. Longer temporal baselines increase the probability of surface change (decorrelation) but capture more displacement.
TOPS	Terrain Observation with Progressive Scans. Sentinel-1's antenna scanning pattern that creates wide-area coverage with uniform image quality but requires ESD coregistration.
Unwrapping	The process of resolving the 2π ambiguity in wrapped interferometric phase to produce continuous absolute displacement values.
VV polarization	Vertical transmit, Vertical receive. The radar sends and receives vertically polarised signals. More sensitive to surface scattering geometry than VH cross-polarization; preferred for subsidence monitoring.
Wavelength (λ)	The distance between consecutive peaks of a radar wave. For Sentinel-1: $\lambda = 0.0554658$ m. Determines the displacement sensitivity: one full phase cycle = $\lambda/2 = 2.77$ cm of LOS displacement.